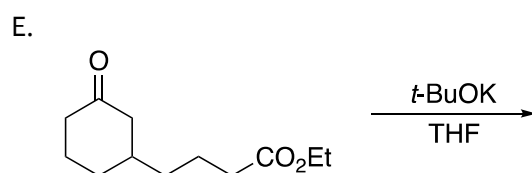
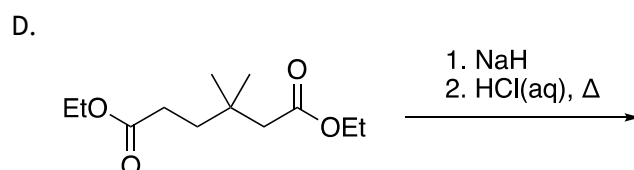
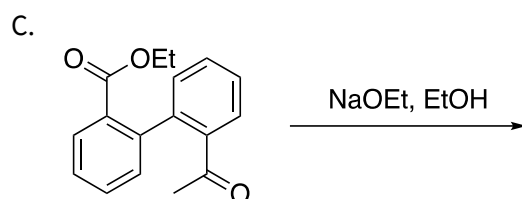
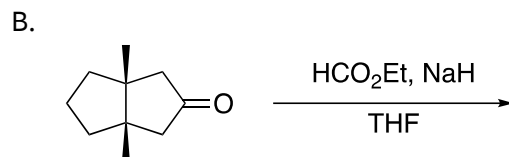
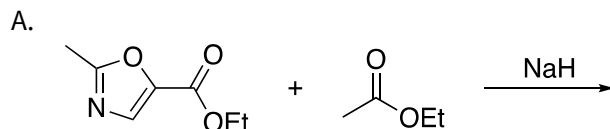
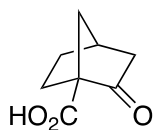


Claisen Condensations and Related Reactions

1. Provide the neutral organic product of each reaction below after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product.

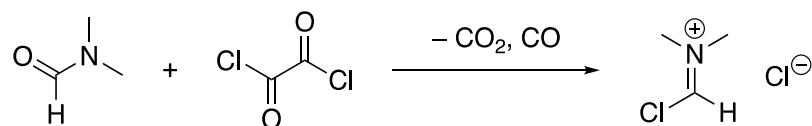


2. Most β -keto carboxylic acids lose carbon dioxide very easily upon gentle heating. The compound below is an exception; it survives heating to very high temperatures without losing CO_2 . Explain why.

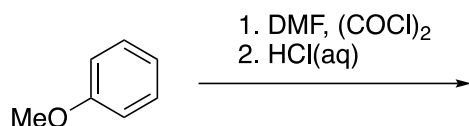


3. The Vilsmeier reaction is a formylation reaction of aromatic compounds with great practical importance because Friedel-Crafts acylation using HCOCl often does not work. The Vilsmeier reagent, $\text{Me}_2\text{NCHCl}^+ \text{Cl}^-$, is an electrophilic source of the formyl group $-\text{CHO}$.

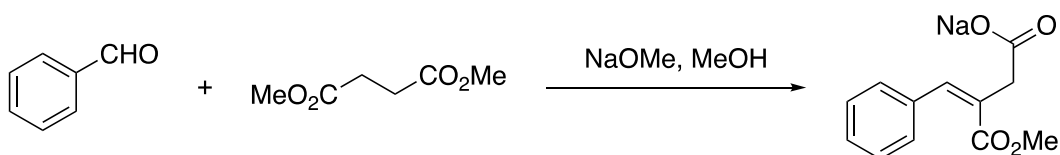
- A. The Vilsmeier reagent can be prepared from dimethylformamide (DMF) and oxalyl chloride. Propose a mechanism for this reaction. List the elementary steps of your mechanism in your response but do not provide curved arrows.



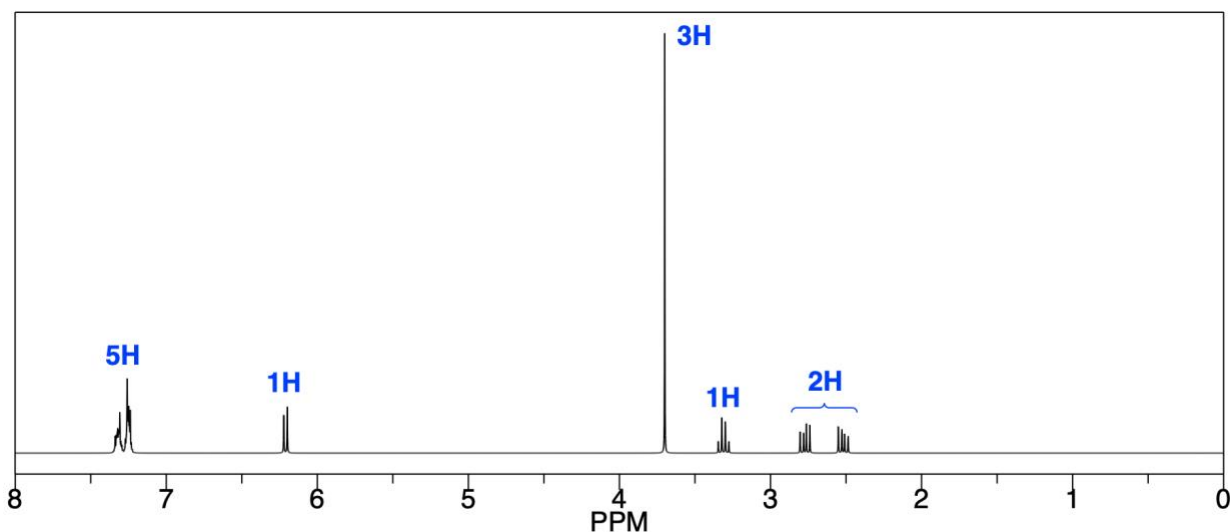
- B. Applying your response to the previous problem and what we've learned about electrophilic aromatic substitutions, draw the major product of the following reaction.



4. Benzaldehyde condenses with dimethyl succinate under basic conditions to afford the unsaturated ester below as a mixture of *E* and *Z* isomers.



The mechanism of this reaction does *not* involve aldol condensation followed by hydrolysis of one of the ester groups. To probe the mechanism, a compound with the ¹H NMR spectrum below was treated with sodium methoxide in methanol and formation of the product was observed, implying that this compound is an intermediate in the mechanism.

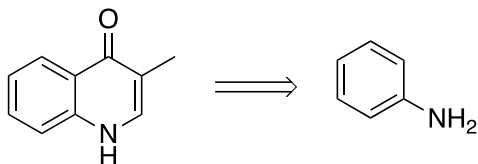


On the basis of this result, propose a curved-arrow mechanism for this reaction.

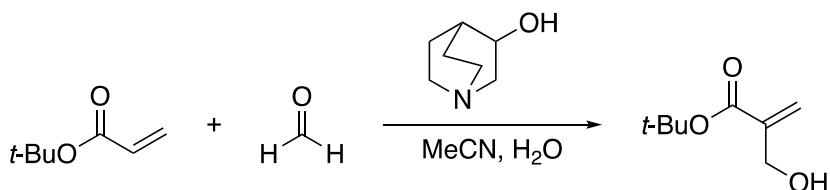
Conjugate Additions and Substitutions

5. Conjugate substitution is a common step in the synthesis of heterocycles. The heterocycle below, a 4-quinolone, can be made from aniline (PhNH_2) through a reaction that combines conjugate substitution and Friedel-Crafts acylation in a single pot. An ester is used as the Friedel-Crafts electrophile instead of an acyl chloride to minimize *N*-acylation.

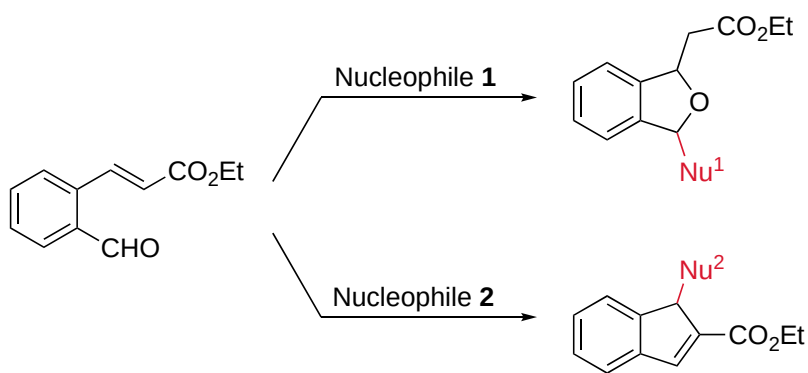
Provide the missing organic substrate that would react with aniline to produce the 4-quinolone shown below.



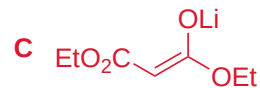
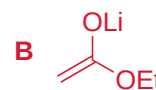
6. Conjugate addition can be used to form enolates. This idea is applied in the Morita-Baylis-Hillman reaction, a kind of nucleophile-catalyzed aldol addition.



- A. Draw the enolate intermediate formed after the best nucleophile in the reaction mixture above engages in conjugate addition.
- B. Propose a curved-arrow mechanism for formation of the product starting from the intermediate you drew in part A. Be sure to regenerate the nucleophilic catalyst.
7. The electrophile shown below was subjected to reaction with the series of nucleophiles listed below in red; one of the two products shown was observed in each case. Predict whether each nucleophile would lead to the top or bottom product and rationalize your predictions.



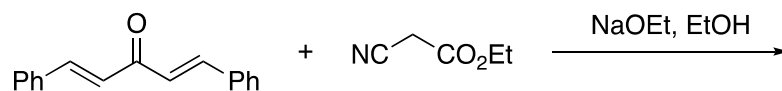
A MeLi, CuCl



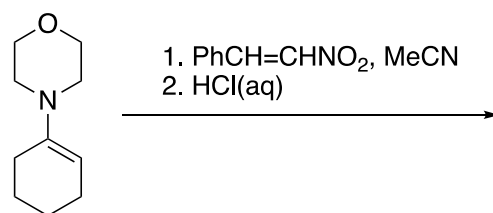
D MeMgBr

8. Provide the neutral organic product of each reaction below after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product.

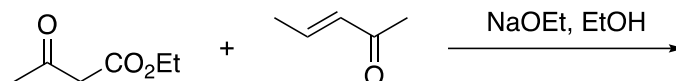
A.



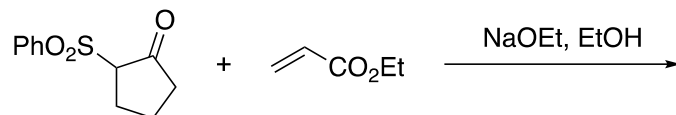
B.



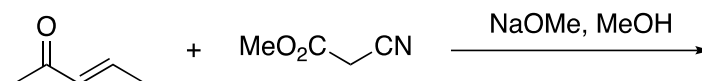
C.



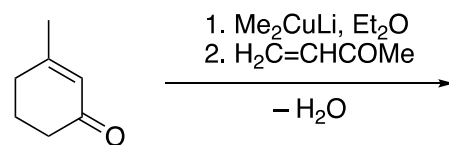
D.



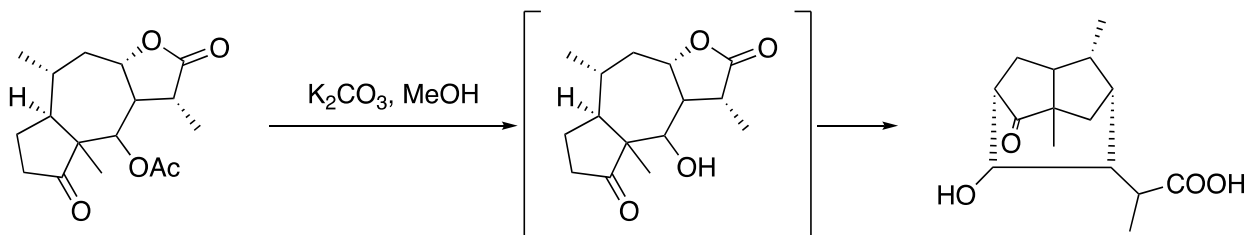
E.



F.

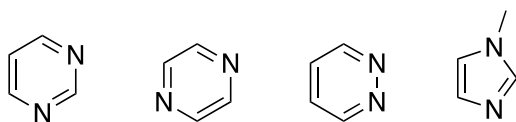
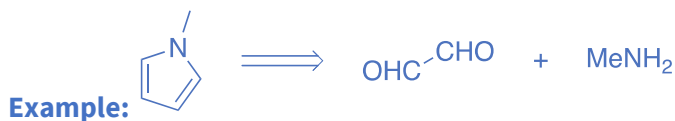


9. The [reaction](#) below involves a retro-aldol process, E1cb elimination, Michael addition, and a final aldol addition. Draw a plausible mechanism with curved arrows for this transformation.



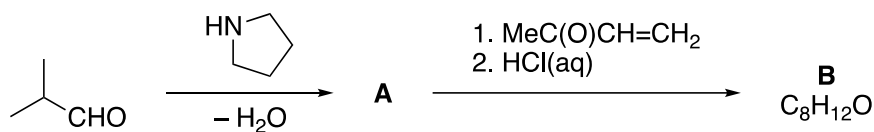
Amines, Imines, Enamines, and Heterocycles

10. Aromatic heterocycles can be made via condensation of a nitrogen-containing nucleophile with a dicarbonyl electrophile. The mechanisms of these reactions resemble imine and enamine formation. For each heterocycle below, provide the structures of a carbonyl compound and nucleophile that could be combined to produce the heterocycle.

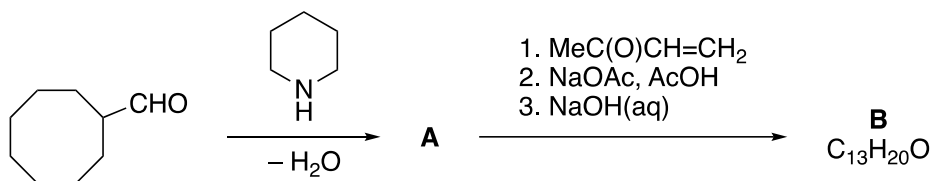


11. Provide the neutral organic product of each reaction below after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product. Provide structures for both **A** and **B** where indicated.

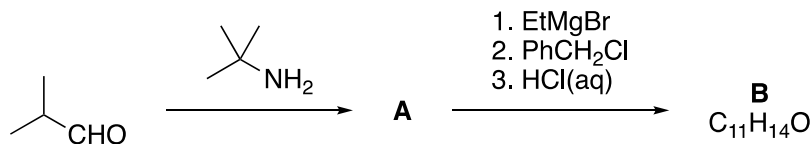
A.



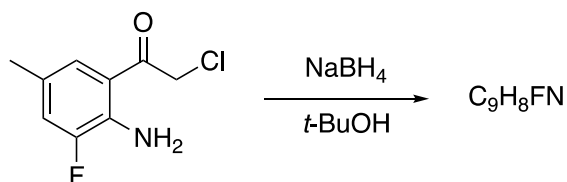
B.



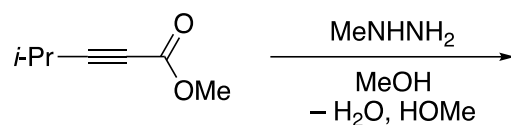
C.



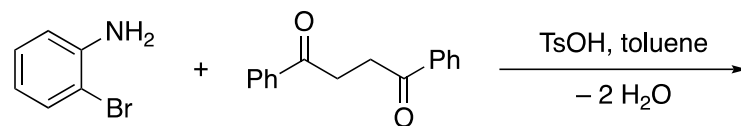
D.



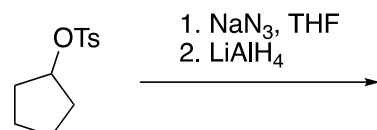
E.



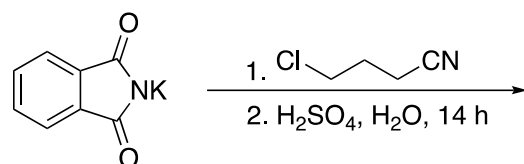
F.



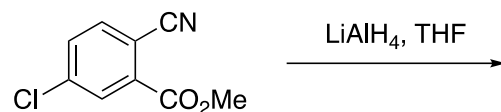
G.



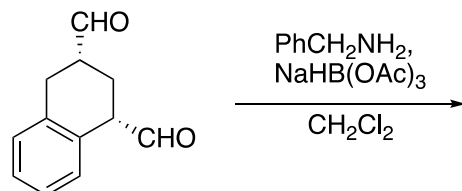
H.



I.

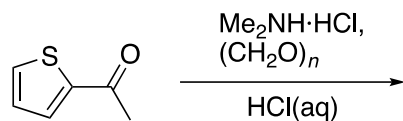


J.

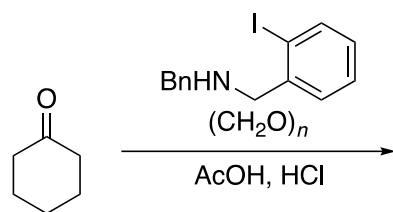


12. For each of the following Mannich reactions, provide the structure of the iminium-containing intermediate formed and draw the structure of the final product.

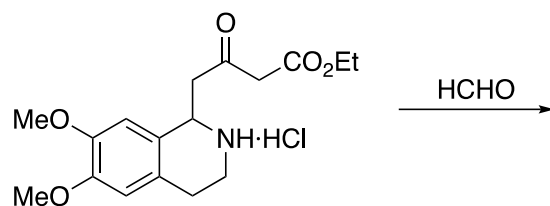
A.



B.

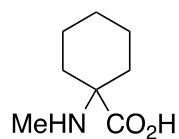


C.

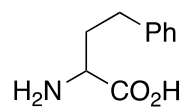


13. The [Strecker reaction](#) is essentially a Mannich reaction with cyanide as the nucleophile. It's an excellent method for the synthesis of non-natural α -amino acids. Design syntheses of the following amino acids starting from a ketone or aldehyde and applying the Strecker reaction.

A.



B.



C.

